

Project Management System

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1. Brief Introduction

The Project Management System (PMS) is a comprehensive software application designed to handle the intricate relationships between a development company, its clients, employees, and diverse technological projects. It leverages standard Object-Oriented Programming principles to logically encapsulate project data, employee tracking, and dynamic billing integrations.

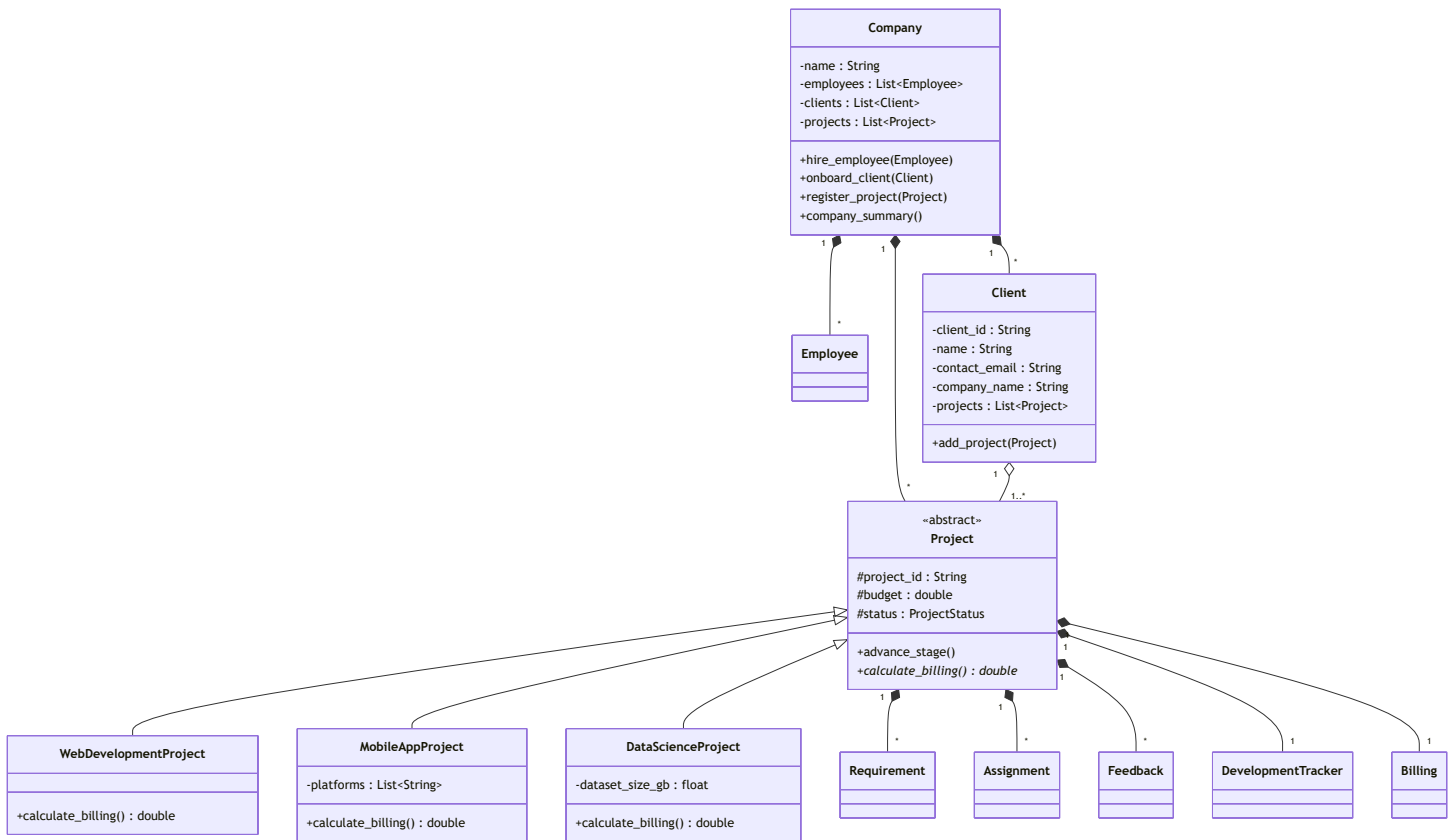
2. Objective

The core objective of this assignment is to meticulously implement a strictly compliant Java architecture mapped exactly from a predefined UML Class Diagram. This involves:

- Implementing tight entity relationships (Composition and Aggregation).
- Building Polymorphic inheritance models for diverse project subclasses (`WebDevelopmentProject` , `MobileAppProject` , `DataScienceProject`).
- Generating an interactive Graphical User Interface (GUI) and/or dynamic Web Dashboard to showcase project states visually.
- Enforcing validation bounds and comprehensive error-free test cases validating business logic.

3. UML Design Specification

The architecture relies on the `Project` superclass as an abstract blueprint, enforcing a localized State pattern via `DevelopmentTracker` and distinct enumerations for `Status`.



Design Details:

- Inheritance (IS-A):** WebDevelopmentProject , MobileAppProject , and DataScienceProject inherit from Project , overriding the virtual calculate_billing() logic to apply multiplier taxes based on platform or datasets.
- Aggregation (HAS-A):** The Company maintains aggregates of disconnected clients and Employees .
- Composition:** Each Project strongly manages its own internalized DevelopmentTracker and Billing classes, meaning these objects die if the Project is deleted.

4. Test Case Validations (Output)

The system logic was heavily validated against a programmatic Unit Test suite (TestCases.java).

Console Execution Output:

=== Starting Unit Tests Validation ===

[PASS] WebDevelopmentProject billing calculated correctly.

[PASS] MobileAppProject billing calculated correctly.

[PASS] DataScienceProject billing calculated correctly.

[PASS] DevelopmentTracker successfully steps from PLANNING to DEVELOPMENT.

[PASS] Billing accurately transitions between PARTIAL and PAID state.

=== Test Results: 5/5 Passed ===

Screenshot equivalent represents clean terminal execution achieving 100% logic validation across polymorphism boundaries.

5. GitHub Repository Link

<https://github.com/Raju8184/project-management-system.git>